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The Palaeomagnetism of Some Igneous Rocks of the Sydney Basin, N.S.W.

E. A. Manwaring

Department of Geophysics, Australian National University, Canberra

Introduction

In the area of the Sydney Basin there are a large number of basic igneous rock bodies whose ages are defined only within wide limits. They are intruded into or extruded upon the Triassic Hawkesbury and Wianamatta Groups. These rocks are thus either Upper- or post-Triassic in age, and have generally been considered as Tertiary (David, 1950). A palaeomagnetic study was made of three of these rock bodies (Fig. 1) with a view to obtaining a more precise estimate of their age.

(1) A dyke of porphyritic olivine dolerite at Luddenham, N.S.W. (33° 53′ S, 150° 40′ E ; map ref. : 703120, Liverpool 1-mile Military, No. 422, Zone 8).

(2) An intrusive (? laccolith) of fine grained olivine dolerite at Peat's Ridge, N.S.W. (33° 19′ S, 151° 12′ E ; map ref. : 217805, Gosford & Norahville 1-mile Military, Nos. 410 & 411, Zone 8).

(3) The basalt flow or flows capping Mt. Tomah in the Blue Mountains of N.S.W. (33° 32′ S, 150° 26′ E ; map ref. : 415550, Katoomba 1-mile Military, No. 415, Zone 8).

This study was begun as part of an Honours thesis in the Department of Geology and Geophysics at the University of Sydney, and completed in the Department of Geophysics, Australian National University, Canberra. I wish to thank Dr. A. A. Day at Sydney, and Mr. E. Irving at Canberra, for much help and advice on this work.

Methods Employed

In each case, samples were distributed as evenly as possible over the total exposures of rock available. At Luddenham, exposure is restricted to three small excavations made some years ago when the rock was quarried as road metal for a short time. Seven oriented samples were taken along about 50 yards of the dyke. At Peat's Ridge, five oriented samples were taken from different places on the working face of the road metal quarry in operation there. On Mt. Tomah, six oriented samples were taken from cuttings on a road (Bell's Line of Road) which crosses the basalt outcrop. Between one and three cylindrical specimens ($1\frac{1}{4}$ inches in diameter ; between $\frac{1}{2}$ and $1\frac{1}{4}$ inches in height) were cut from each oriented sample.

The intensities and directions of the natural remanent magnetization (N.R.M.) of the specimens were measured firstly on the astatic magnetometer in the Department of Geology and Geophysics, Sydney University, in 1960, and later on an astatic magnetometer in the Department of Geophysics, Australian National University, in 1961–62. The stability of the specimens was studied by demagnetization in alternating magnetic fields (Thellier and Rimbert, 1954 and 1955), using the apparatus and procedures described by Irving, Stott and

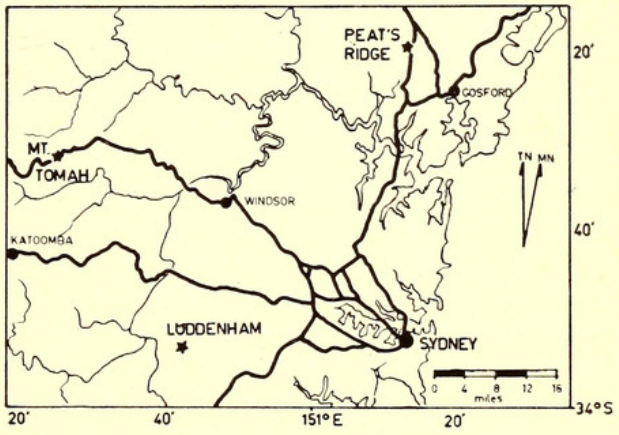


Fig. 1

Sketch map of Sydney area—sampling sites indicated by stars

TABLE 1

Average directions of magnetization and statistical data for Luddenham dolerite dyke and Peat's Ridge dolerite intrusion

S=number of samples ; N=number of specimens ; D=declination of average direction ; I=inclination of average direction ; R_s=vector resultant of sample means ; R_N=vector resultant of specimen directions ; k_w=within-sample precision ; k_b=between-sample precision ; k_o=over-all precision ; a=half-angle of cone of 95% confidence (P=0·05)

	Luddenham			Peat's Ridge		
	NRM Sydney	NRM Canberra	Demag. 225 oe.	NRM Sydney	NRM Canberra	Demag. 225 oe.
S	7	7	7	5	5	5
N	15	15	15	10	10	10
D	117°	117°	314°	172°	175°	190°
I	−84°	−84°	−79°	+70°	+66°	+69°
R_s	5·70	5·86	6·72	4·97	4·95	4·96
R_N	12·33	12·19	14·06	9·93	9·86	9·92
k_w	294	25	27	1500	147	362
k_b	5	6	30	128	100	132
k_o	34	36	138	614	373	558
a	24°	24°	12°	6°	7°	6°

Ward (1961). The method adopted was to select a single specimen from each of a number of samples (from four to six) from each of the rock bodies ; these sets of specimens are defined as the "pilot" specimens. These pilot specimens were then treated in progressively stronger alternating magnetic fields, their magnetizations being measured after each treatment. The alternating field strength ($H_p\sim$) quoted is always the peak value. For each set of pilot specimens the variation of the precision of the directions with treatment was studied.

The changes in the intensity of magnetization of the pilot specimens with progressive partial demagnetization are plotted as a graph of the ratio M/M_o against the alternating field ($H_p\sim$), where M_o is the initial (N.R.M.) intensity of the specimen, and M is the intensity remaining after treatment.

The directions were analysed statistically using Fisher's treatment of dispersion on a sphere (Fisher, 1953). The precision used is k, the estimate of Fisher's $\varkappa$. In order to study the precision of the directions within- and between-samples for the same rock unit, the two-tier statistical analysis given by Watson and Irving (1957) was used. The estimated precision of the specimen directions from samples (within-samples precision) is denoted by k_w ; the between-sample precision by k_b ; and the over-all precision by k_o. The half-angle (a) of the cone of 95% confidence is calculated from k_o. This rather sophisticated analysis was carried out in order to make comparisons between the Sydney and Canberra measurements.

All directions of magnetization are plotted on polar stereographic projections (Wulff net). Negative inclinations (N pole up) are plotted as circles on the upper hemisphere of the projection ; positive inclinations (N pole down) are plotted as dots on the lower hemisphere. Pole positions are plotted on a Lambert polar equal-area projection (Schmidt net) of the Earth's southern hemisphere.

Results from Luddenham Dyke

The directions of the N.R.M. of the Luddenham specimens as measured in (*a*) Sydney, and (*b*) Canberra are plotted in Fig. 2. A comparison shows that the two separate measurements for individual specimens do not, in most cases, agree. This is particularly evident in the directions in the SE quadrant.

A statistical comparison of the results from the two sets of readings is given in Table 1. The average direction for both sets is exactly

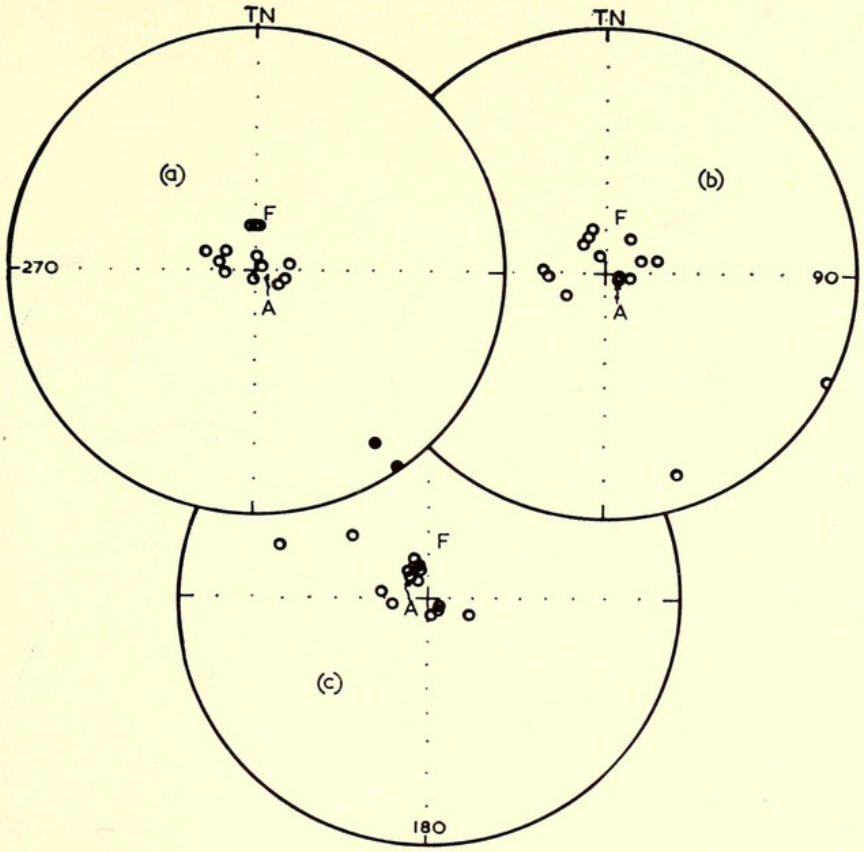


FIG. 2

Directions of magnetization of the specimens from Luddenham dyke

(*a*) N.R.M. directions as measured in Sydney in 1960
(*b*) N.R.M. directions as measured in Canberra in 1961–62
(*c*) Directions of magnetization after treatment in an alternating magnetic field of 225 oersteds (peak)

The average direction is indicated by " A " and the direction of the present Earth's field by " F "

the same, and the between-sample precision and over-all precision of each are very similar. Yet the within-sample precision is much less for the Canberra readings than for the Sydney ones. This could be due to the Canberra readings being of lower accuracy, but repeat measurements on specimens at *both* Sydney and Canberra give results whose standard deviation is approximately 2°, which is too small to explain the differences observed in Fig. 2. It would appear therefore that the scatter of the individual specimen directions has increased in the period of one year between measurements. This shows an instability of the magnetizations of the specimens which would be due to a randomly directed component of viscous magnetization imposed on them during this period (the specimens were stored in random positions in the earth's field).

In Fig. 3 are plotted the changes in the directions of magnetism of the six pilot specimens as the alternating field strength was increased. The variation of the precision between the directions of these specimens is plotted in Fig. 4. It is apparent that the greatest precision is obtained after partial demagnetization in a field of 225 oersteds (peak). The other specimens from Luddenham were treated in this field, and their directions are plotted in Fig. 2 (*c*). The average direction and statistical data are given in Table 1. It can be seen that there is little increase in the within-sample precision, but that there is a large increase in the between-sample precision and in the over-all precision.

There are two methods by which the optimum partial demagnetization field may be chosen, and it is of interest to compare them. Firstly, there is the method used above. Secondly, there is the method used by As and Zijderveld (1958), in which the changes in the directions of specimens are studied individually. These workers found that with progressive demagnetization the unstable component of magnetization diminishes and even vanishes without greatly affecting the direction of the original stable magnetization. If the unstable component

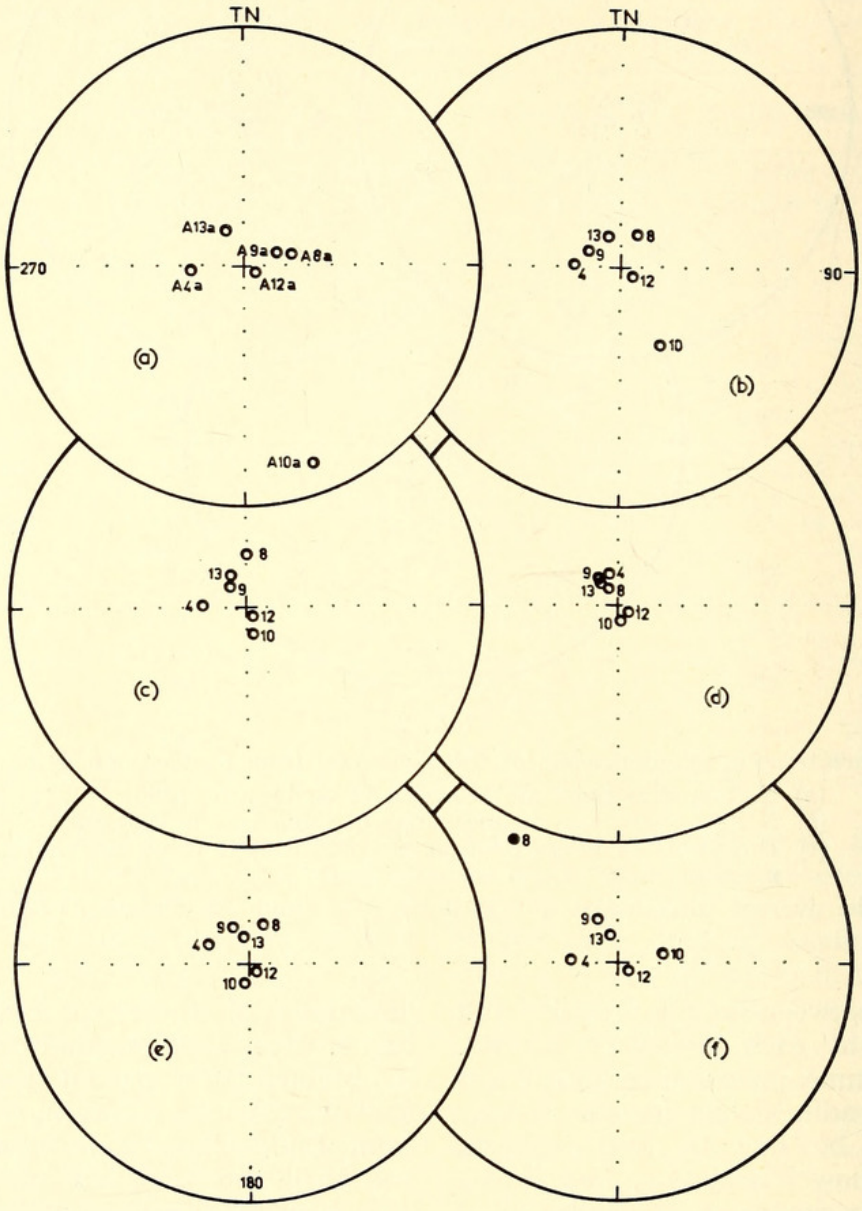


FIG. 3

The changes in the directions of magnetization of the Luddenham pilot specimens (A4a, A8a, A10a, A12a and A13a) with treatment in alternating fields

The directions plotted are :

(*a*) N.R.M.'s
(*b*) After treatment in 75 oersteds (peak)
(*c*) After 150 oe.
(*d*) After 225 oe.
(*e*) After 300 oe.
(*f*) After 375 oe.

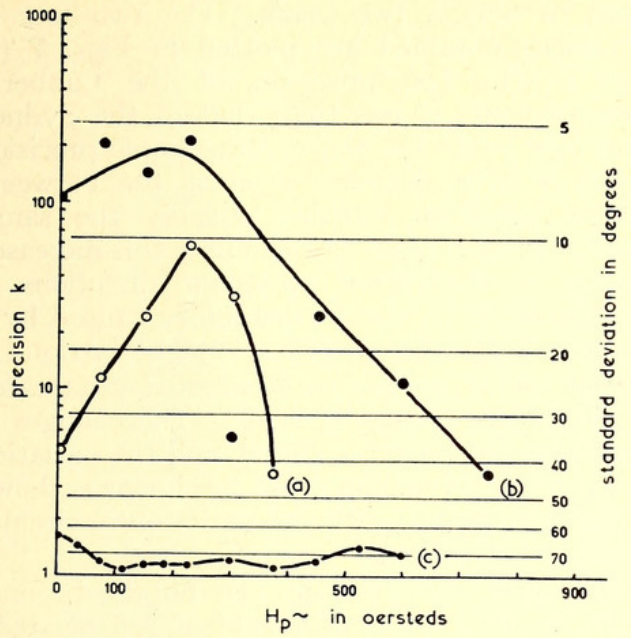


FIG. 4

The variation of the precision k (Fisher, 1953) of the directions of the pilot specimens from (*a*) Luddenham, (*b*) Peat's Ridge, and (*c*) Mt. Tomah, with treatment in alternating fields ($H_p\sim$). The precision k is plotted on a logarithmic scale.

Also shown are the values of k at which the standard (circular) deviation (Creer, Irving and Nairn, 1959) has values as indicated on the figure. Thus the variation of this circular standard deviation with $H_p\sim$ can also be seen

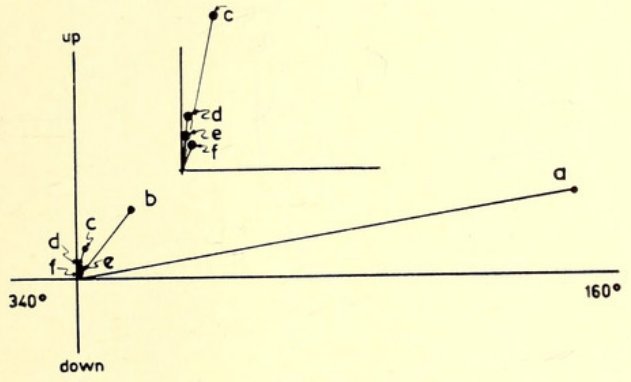


FIG. 5

Vector plot of the changes in the directions of specimen A10a with treatment in alternating fields. The vectors are plotted in the vertical plane which lies in the azimuth of 160° (true). The lengths of the vectors are proportional to the remaining intensity of magnetization after each treatment, and the angles of inclination are plotted in the vertical plane. The alternating field strengths and intensities of magnetization (M) are as follows: (*a*) N.R.M.; $M=191\times10^{-4}$ emu/cc; (*b*) 75 oe; $M=33\times10^{-4}$ emu/cc; (*c*) 150 oe.; $M=11\cdot5\times10^{-4}$ emu/cc; (*d*) 225 oe.; $M=3\cdot8\times10^{-4}$ emu/cc; (*e*) 300 oe.; $M=2\cdot5\times10^{-4}$ emu/cc; (*f*) 375 oe.; $M=1\cdot9\times10^{-4}$ emu/cc. Inset are shown the vectors *c*, *d*, *e* and *f*, to a larger scale

is at some angle to the direction of the stable one, the direction of the total magnetization rotates with demagnetization, until a stage is reached where the resulting magnetization remains constant in direction and decreases in magnitude. It is then assumed that the unstable component has been completely removed, and that the constant direction obtained is that of the original, stable magnetization of the rock. The final constant directions of all the specimens are then combined to give the average result.

This process of obtaining a final, relatively constant direction can be seen in Fig. 5, where the changes in direction and intensity of specimen A10a are plotted as vectors. (A10a is the only specimen which has a sufficiently large change in direction for such a figure to be drawn.) It can be seen from this figure that the field strength chosen for treating the remainder of the Luddenham specimens (225 oe.) gives a direction which lies within the range of this final, constant direction. Thus the two methods agree in this particular case.

The manner in which the intensities of magnetization (moments) of the pilot specimens change is shown in Fig. 6 by a plot of M/M_o against $H_p\sim$: the values of M/M_o at 225 oe. are compared with the total angular change in direction of magnetization for each specimen in Table 2. It is apparent that, in a general way, the smaller the ratio—i.e. the greater the relative decrease in moment—then the less the stability—i.e. the greater the change in direction. Thus, referring to Fig. 6, specimen A10a is the least stable and specimens A12a and A13a the most stable.

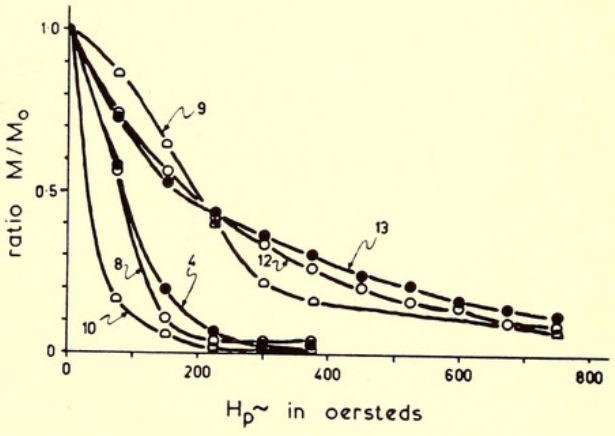


FIG. 6

The variation of the ratio M/M_o with alternating field strength ($H_p\sim$) for the Luddenham pilot specimens. M_o is the initial (N.R.M.) intensity of magnetization, and M the intensity after treatment

TABLE 2

Comparison of the total angular change in the directions of magnetization of the pilot specimens from Luddenham (after treatment at 225 oersteds) with the ratio of the intensity remaining (M) to the initial (N.R.M.) intensity (M_o)

Specimen	M/M_o (225 oe.)	Angular Change
A10a	0·02	74°
A8a	0·04	28°
A4a	0·07	27°
A9a	0·41	25°
A12a	0·43	1°
A13a	0·43	8°

Results from the Peat's Ridge Dolerite

The magnetizations of the Peat's Ridge specimens differ from the Luddenham ones in that they are all reversely magnetized (inclinations positive), and that the precision of their N.R.M. directions is greater.

The N.R.M.'s were measured initially in Sydney, and then again in Canberra after a lapse of almost two years. The two sets of directions obtained are plotted in Figs. 7 (*a*) and 7 (*b*). The precision of the Canberra measurements is less than that of the Sydney ones (see Table 1)—the within-sample precision shows a large decrease whereas the between-sample precision remains sensibly the same. For the same reasons given before, this increased scatter of individual specimen directions is considered to be due to instability caused by a random component of viscous magnetization.

Four pilot specimens were selected and treated in alternating magnetic fields. The changes in directions are shown in Fig. 8, and the variation of the precision among these specimens is shown in Fig. 4, together with the variation in circular standard deviation.

The greatest precision is obtained, once again, by treatment in a field of 225 oersteds. The directions of all the Peat's Ridge specimens after treatment are shown in Fig. 7 (*c*), and the average direction and statistical data are given in Table 1.

In this case, the within-sample precision is increased, but does not regain the value it had

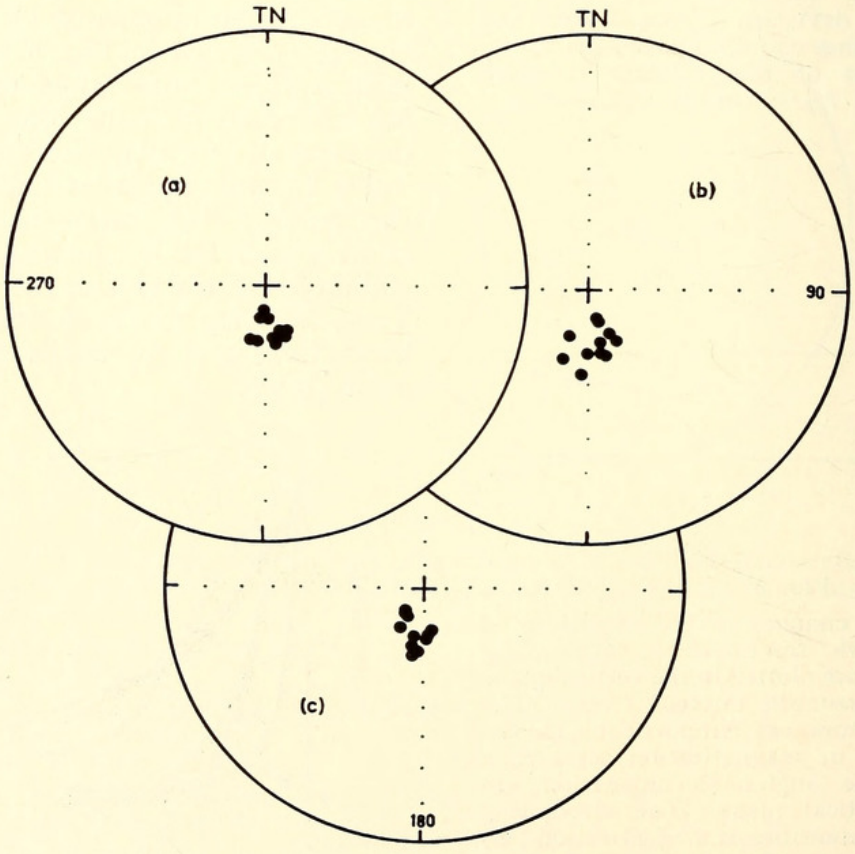


FIG. 7

Directions of magnetization of the specimens from the Peat's Ridge intrusion

(*a*) N.R.M. directions measured in Sydney
(*b*) N.R.M. directions measured in Canberra
(*c*) Directions after treatment in an alternating field of 225 oersteds

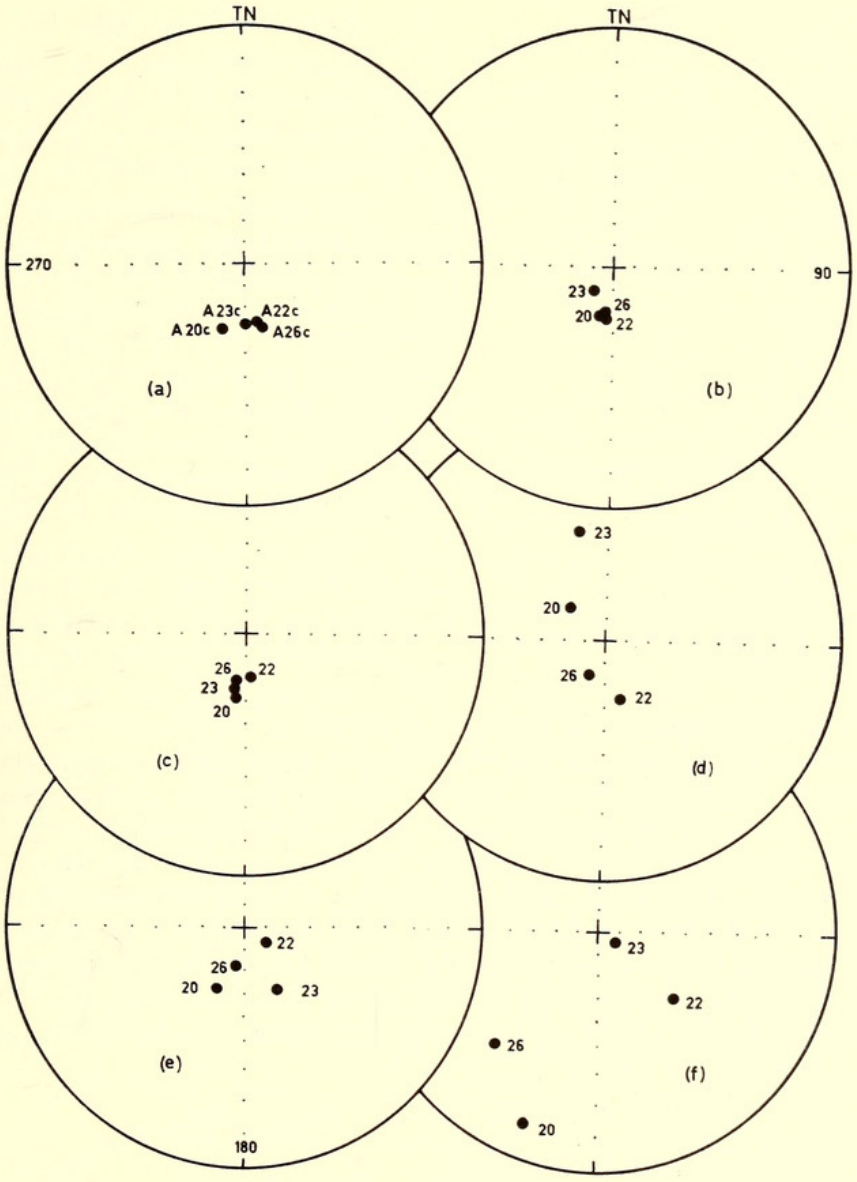


FIG. 8

The changes in the directions of the Peat's Ridge pilot specimens (A20c, A22c, A23c and A26c) with treatment in alternating magnetic fields. The directions plotted are :

(*a*) N.R.M.'s
(*b*) After treatment in 150 oersteds (peak)
(*c*) After 225 oe.
(*d*) After 300 oe.
(*e*) After 450 oe.
(*f*) After 750 oe.

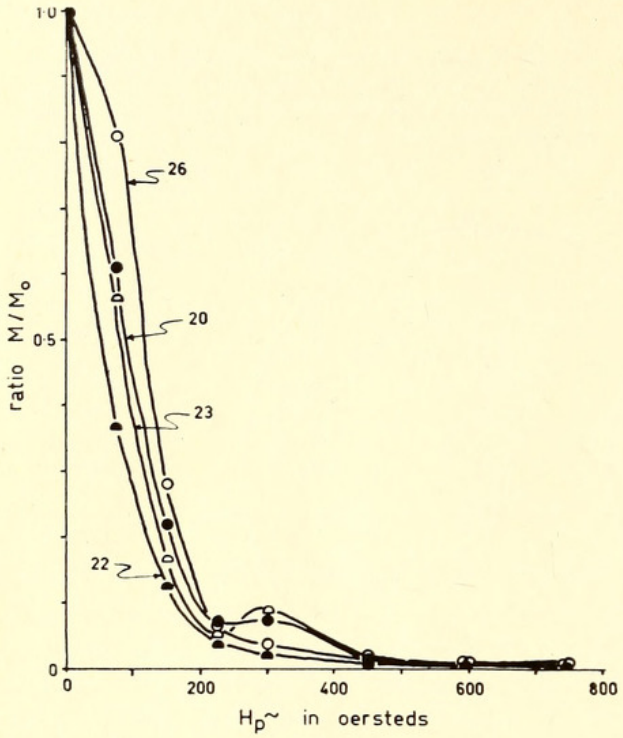


FIG. 9

The variation of the ratio M/M_o with alternating field strength ($H_p\sim$) for the Peat's Ridge pilot specimens

when the N.R.M.'s were measured in Sydney. On the other hand, the between-sample precision increased to a value slightly greater than originally. And so the over-all precision is not appreciably less than that of the Sydney measurements. It would thus seem that the greater part of the random viscous magnetization imposed in the period between the N.R.M. measurements was removed by the treatment in alternating magnetic fields. Also, since the average direction has changed, a common and more-or-less constant unstable component has been removed from all specimens.

The ratio M/M_o is plotted against $H_p\sim$ in Fig. 9 for the pilot specimens.

Results from the Mt. Tomah Basalts

Both the Sydney and Canberra measurements of the N.R.M.'s of the Mt. Tomah specimens gave a very large scatter of directions. These directions are plotted in Figs. 10 (*a*) and 10 (*b*).

Six pilot specimens were selected and treated in alternating magnetic fields. The changes in their directions are shown in Fig. 11, and the variation of the precision of these directions is shown in Fig. 4. The precision remains very low at all stages of the treatment. There is no optimum treatment field indicated by the curve in Fig. 4, as was the case for both Luddenham and Peat's Ridge, and so the remainder of the specimens were not treated at all.

The variation of M/M_o for Mt. Tomah is plotted in Fig. 12, and shows a very swift decrease in the intensities of the pilot specimens (compare with Figs. 6 and 9).

On applying the randomness test of Watson (1956) to the directions obtained from the pilot specimens during treatment, it was found that the distribution of directions was random (at the 5% level of significance) initially and at all stages of the treatment. The ratio k/k_{sig} (k_{sig} is the minimum value of k for the distribution of directions to be significant, as calculated from the tables of the significance points of R given by Watson) varied between 0·7 and 0·5. The cause of this wide scatter is not known,

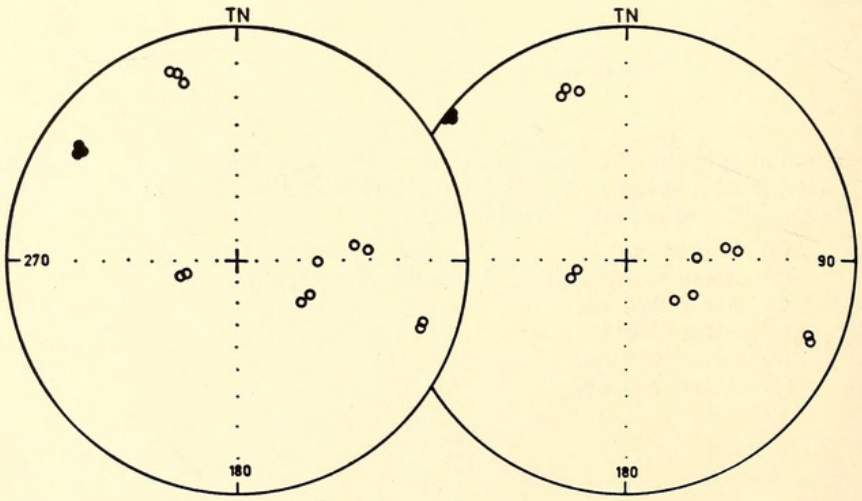


FIG. 10

Directions of magnetization of the specimens from the Mt. Tomah basalts

(*a*) N.R.M. directions as measured in Sydney
(*b*) N.R.M. directions as measured in Canberra

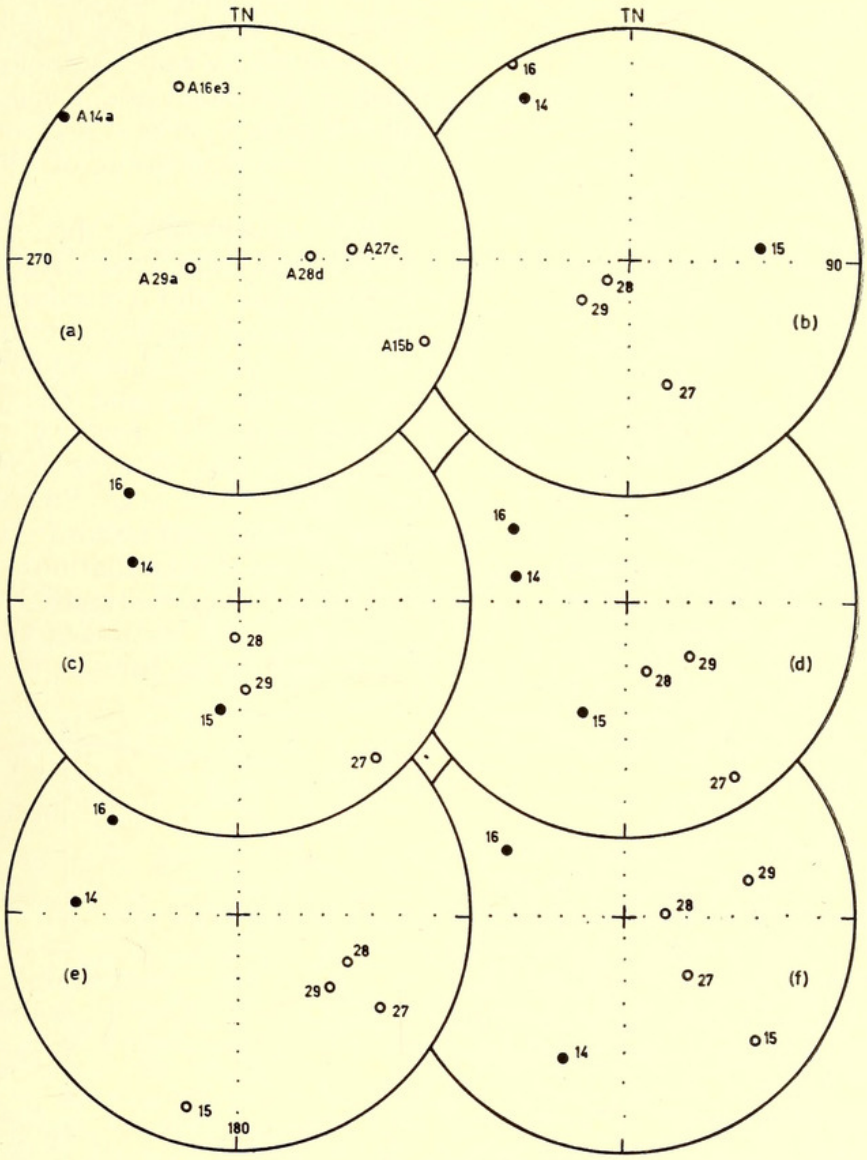


FIG. 11

The changes in the directions of magnetization of the Mt. Tomah pilot specimens (A14a, A15b, A16e3, A27c, A28d and A29a) with treatment in alternating magnetic fields. The directions plotted are:

(*a*) N.R.M.'s
(*b*) After treatment in 75 oersteds (peak)
(*c*) After 150 oe.
(*d*) After 300 oe.
(*e*) After 450 oe.
(*f*) After 600 oe.

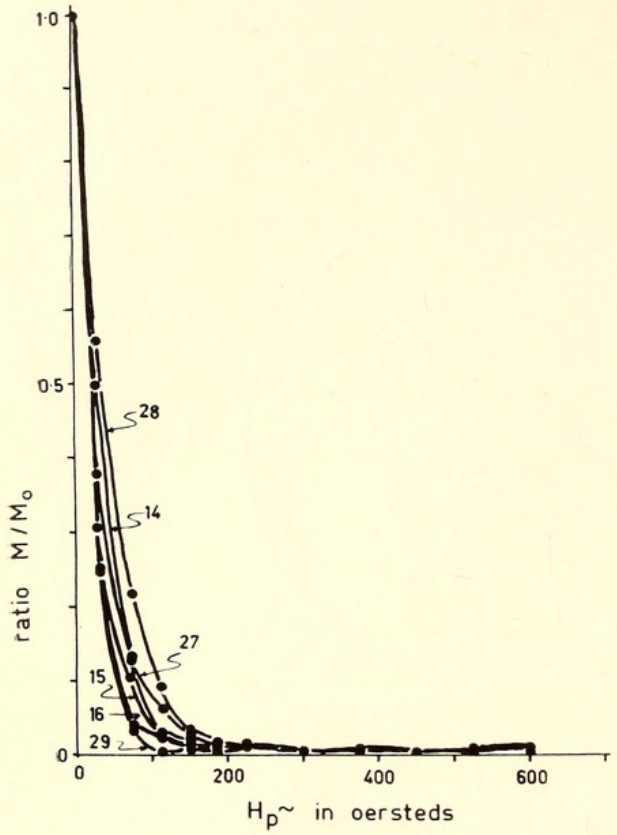


FIG. 12

The variation of the ratio M/M_o with alternating field strength ($H_p\sim$) for the Mt. Tomah pilot specimens

but it is clearly not due to viscous components of magnetization or to lightning strikes alone, as these would be removed at least partially by the treatment. It could possibly be due to the original thermo-remanent magnetization decaying away completely with time, leaving only randomly directed magnetizations behind.

Discussion

The directions of magnetization of Luddenham and Peat's Ridge after treatment in an alternating magnetic field of 225 oersteds (peak) are stable, and their average directions may be considered to be coincident with the directions of the geomagnetic field at the time these bodies cooled. For a comparison between these, and for comparison with the results obtained by other workers on similar bodies, pole positions were calculated from these average directions, assuming the earth's field to be a geocentric dipole. A comparison of pole positions eliminates differences caused by differences in the geographic positions of sampling sites. Some error is most probably present in the pole positions calculated, since the secular variation of the earth's field is probably not averaged out in the samples taken, due to the comparative shortness of the time the bodies would take to cool.

The Luddenham and Peat's Ridge southern hemisphere pole positions are given in Table 3, together with the poles of those rock bodies with which they are compared. Also given in this table are the ages of four of these bodies as determined by radio-isotopic means (McDougall, 1961 ; Evernden and Richards, 1962). A plot of the pole positions is given in Fig. 13, on an equal-area projection of the earth's southern hemisphere.

By considering the pole positions of Luddenham and Peat's Ridge in relation to the other poles, and considering the geologically and radio-isotopically determined ages, it seems that the dyke at Luddenham is most probably Mesozoic in age, and the intrusion at Peat's Ridge probably Tertiary. It is noteworthy that estimates of ages from previous palaeomagnetic work (e.g. Prospect and Gibraltar) have been proved accurate by subsequent radio-isotopic age determinations. Both Prospect and Gibraltar were estimated as being in the Mesozoic, and radio-isotope work has shown them both to be Jurassic (see Table 3).

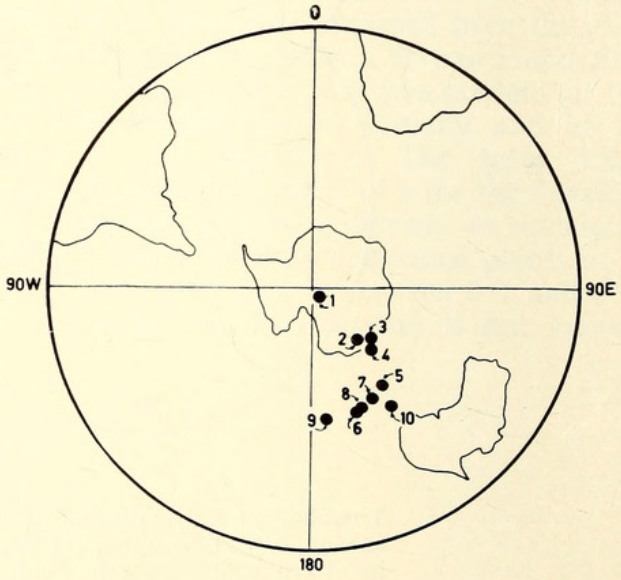


FIG. 13

Equal-area stereographic projection of the Earth's southern hemisphere, on which are plotted the pole positions of the rock bodies from south-eastern Australia given in Table 3. The pole positions are numbered as follows :

1. Newer volcanics of Victoria
2. Peat's Ridge dolerite
3. Older volcanics of Victoria
4. Tertiary basalts of N.S.W.
5. Gingenbullen dolerite
6. Cygnet alkaline complex
7. Prospect dolerite
8. Tasmanian dolerites
9. Luddenham dolerite
10. Gibraltar syenite

TABLE 3

Comparison of the palaeomagnetic pole positions and geological ages of igneous rock bodies of south-eastern Australia

References :

1. Boesen, Irving and Robertson (1961)
2. Irving (1963)
3. Irving and Green (1957)
4. Irving, Stott and Ward (1961)
5. Robertson and Hastie (1962)
6. Robertson (1963)

The radio-isotopic age of the Tasmanian dolerites is taken from McDougall (1961), and the rest from Evernden and Richards (1962)

Rock Unit	Ref.	Geological Age	Pole Position	Radio-isotopic Age in millions of years
Gibraltar Syenite	1	post-Upper Triassic	41S, 146E	178 Lower Jurassic
Luddenham Dolerite		post-Upper Triassic	47S, 173E	
Tasmanian Dolerite	2	post-Upper Triassic and pre-Tertiary	51S, 160E	167 Middle Jurassic
Prospect Dolerite	1	post-Upper Triassic	51S, 151E	168 Middle Jurassic
Cygnet Complex	5	post-Tasmanian Dolerites	50S, 158E	104 Lower Cretaceous
Gingenbullen Dolerite	1	post-Upper Triassic	53S, 144E	
Mt. Dromedary Complex	6	post-Permian	56S, 138E	Middle Cretaceous 93
Tertiary Basalts of N.S.W.	4	Tertiary	63S, 137E	
Older Volcanics of Victoria	3	Lower Tertiary	67S, 123E	
Peat's Ridge Dolerite		post-Middle Triassic	70S, 134E	
Newer Volcanics of Victoria	3	Pliocene to Recent	86S, 102E	

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